

37.9 A 14Gb/s/pin 0.163pJ/b DQ Receiver for HBM with Baud-Rate Phase Tracking Loop Supporting Background Offset Calibration

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This paper presents a compact 14Gb/s/pin HBM DQ receiver that continuously tracks sampling phase and detector offset in live data, using a baud-rate phase tracking loop w... [Show More](#)

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The shift in high-performance computing and memory systems toward multi-chip module packaging, short-reach die-to-die links, and standardized chiplet interfaces is driving requirements for increased bandwidth and I/O density at a low energy per bit [1]– [6]. Unlike long-reach serial I/Os, high-density memory interfaces typically avoid clock and data recovery and instead rely on source-synchronous clocking and periodic training to save area and power [7]– [10]. However, as high-bandwidth memories (HBM) scale to thousands of I/O lanes, PVT drift, aging, SSN/PSIJ, and package/board coupling cause lane-to-lane variations and time-varying offsets. These effects cannot be fully suppressed by initial training alone [11]– [14]. As a result, HBM DQ links face three practical challenges: (1) perlane V_{REF} generation and trimming, and comparator-offset cancellation consume a large area and do not address time-dependent residual offsets; (2) the DQS-DQ phase relationship drifts while live data is transmitted; thus, reducing eye margins if not continuously tracked; and (3) the dense channel pitch exacerbates crosstalk while simultaneous-switching noise results in supply-induced margin losses for single-ended links. The third challenge is primarily addressed by signaling and packaging choices and is not the focus of this work. To address these challenges, prior work employed V_{REF} tracking or frequent retraining [3], [8] [10]. V_{REF} -tracking loops often have limited bandwidth due to a complex feedback loop; whereas, retraining interrupts live data transmission and becomes increasingly costly in wide interfaces, where thousands of lanes must periodically suspend traffic for training. Motivated by these limitations, this work focuses on challenges (1) and (2) - offset wander and phase drift during live data transmission - and introduces a baud-rate phase-tracking loop with background offset calibration that continuously aligns the sampling phase and detector threshold on live data.



[Figure 37.9.1](#) shows the overall block diagram of the proposed DQ receiver, with a baudrate phase-tracking loop and background offset-calibration logic. D_{in} represents the 14 Gb/s single-ended data input, and CLK_{in} is the 7 GHz differential clock inputs. A clock divider (CLK DIV) converts the half-rate clock into quarter-rate 4-phase clocks, CLK_0 , CLK_{90} , CLK_{180} , and CLK_{270} ; skew between the 4-phase clocks is manually adjusted. A digitally controlled delay line (DCDL) adjusts the quarter-rate clock delays via a 6b DCDL control word (DCW) to achieve an optimal sampling point. Fine phase adjustment is performed by a time-window-based phase detector (TWPd) that integrates D_{in} over the interval between the rising edges of CLK_0 and CLK_{90} and generates a 1-bit digital output T by comparing the resulting integrated voltage with an internal decision threshold. This threshold is defined as the effective input level at which the differential TWPd outputs, V_{OUTP} and V_{OUTN} , become equal; in the absence of internal offset, it coincides with the DC reference level V_{REF} of D_{in} . However, internal TWPd offsets due to process and temperature variation result in threshold voltage variation. To explicitly identify data transitions on D_{in} within the CLK_0 - CLK_{90} window, rising-edge-validation (REV) and falling-edge-validation (FEV) circuits generate 1-bit flags R and F , respectively: $R = 1$ ($F = 1$) indicates that a $0 \rightarrow 1$ ($1 \rightarrow 0$) transition occurred between the two clock edges, while $R = F = 0$ indicates no transition in the window. For an ideal, offset-free TWPd, using only R would be sufficient for phase tracking; however, observing both R and F enables the background offset-calibration logic to distinguish true sampling-phase error from detector offset by checking whether T changes sign with transition polarity. The digital control logic therefore processes the patterns of (R, F, T) and, depending on whether T behaves consistently with phase error or appears stuck high/low due to offset, adjusts either the calibration-control word (CCW) to cancel TWPd offset or the DCW to realign the sampling phase.



Figure 37.9.1

Proposed DQ receiver block diagram, showing the baud-rate phasetracking loop and background offset-calibration path.

[Figure 37.9.2](#) illustrates TWPd operation and possible outputs for different offset conditions. In the TWPd circuit of [Fig. 37.9.1](#), C_{Din} denotes the capacitance connected to D_{in} and is adjusted by CCW, while C_{REF} denotes the capacitance connected to the reference node REF. When no data transition occurs within the CLK_0 - CLK_{90} integration window, the TWPd output T simply indicates whether D_{in} is above or below the internal decision threshold over that interval, producing a binary value of 0 or 1 accordingly. When a data transition occurs inside the window and no detector offset is present ($C_{Din} = C_{REF}$), T becomes a function of the sampling-clock phase relative to D_{in} . If the sampling clock lags D_{in} and a rising transition occurs within the window, D_{in} spends more time above the threshold than below it over the integration interval, so T resolves to 1; for a falling transition under the same lagging condition, D_{in} spends more time below the threshold and T resolves to 0. When CLK leads D_{in} , then $T=0$ if the rising transition occurs since D_{in} spends more time low than high; conversely, $T=1$ if a falling transition occurs. Thus, under offset-free conditions the combination of R , F , and T reveal the phase offset polarity; thus, enabling precise DCW adjustment. On the other hand, when an offset is present, the TWPd output is influenced by offset. If C_{Din} is smaller than C_{REF} , T is likely to remain at 0 irrespective of whether a transition occurs earlier or later than the optimal point. Conversely, if C_{Din} is larger than C_{REF} , T is likely to remain at 1 regardless of the transition timing. In this regime, the observed combinations of R , F , and T deviate from those expected for pure phase error and instead reveal the polarity of the detector offset. The calibration logic interprets these biased patterns to iteratively adjust CCW until $C_{Din} \approx C_{REF}$ and T recovers its expected dependence on transition polarity and clock phase, after which the phase-tracking loop can use the same signals to refine DCW.



Figure 37.9.2

Signal diagrams demonstrating TWPd operation for rising and falling data transitions under three detector-offset conditions: $C_{Din} = C_{REF}$, $C_{Din} < C_{REF}$, and $C_{Din} > C_{REF}$.

[Figure 37.9.3](#) details the background offset calibration logic using REV, FEV, and TWPd, interpreted in the voltage variation of D_{in} and the TWPd threshold. The panels plot samplingclock lock points and rise/fall early/late scenarios for three conditions: $C_{Din} = C_{REF}$, $C_{Din} < C_{REF}$, and $C_{Din} > C_{REF}$. When

$C_{Din} = C_{REF}$, the TWPd threshold aligns with the common level of D_{in} ; a nominal lock occurs where $A = B$ and $C = D$, and rise/fall decisions of T are complementary. On the other hand, if the threshold is displaced by offset, collapsing the rise/fall decisions and biasing the early/late classification even when the sampling phase is positioned at the offset-free lock point. Specifically, for $C_{Din} < C_{REF}$ (threshold shifted high), a rise transition is misclassified as late and a fall transition as early at the $C_{Din} = C_{REF}$ lock point; for $C_{Din} > C_{REF}$ (threshold shifted low), a rise is misclassified as early and a fall as late. Near this point, T tends to take the same value for both transition polarities, providing a distinct marker of detector offset. Consequently, the combinations of R , F , and T becomes inconsistent with pure phase error and instead encodes the offset polarity; the synthesized digital logic exploits these signatures to adjust CCW until rise/fall symmetry – and correct early/late labeling – is restored.

Figure 37.9.3

Background offset-calibration logic operation, using REV, FEV, and TWPd from the perspective of voltage variation.

[Figure 37.9.4](#) presents the simulated phase tracking behavior for initial optimal clock phase and demonstrates the effectiveness of background offset calibration. In the simulations, three offset conditions are imposed in the TWPd: offset 1 from 0 to 100 ns, offset 2 from 100 ns to 300 ns, and offset 3 from 300 ns to 600 ns. As a result, with background offset calibration, offset cancellation is achieved by adjusting CCW to 8, 15, and 0 for offsets 1, 2, and 3, respectively, without introducing phase offsets. Therefore, the DCW lock point remains fixed at 30, corresponding to the optimal sampling point. On the other hand, without background offset calibration, the TWPd fails to track offset transitions, causing deviations from the optimal lock point.

Figure 37.9.4

Simulated phase tracking behavior for an initial optimal clock phase and the effect of background offset calibration.

The proposed 4-lane DQ receiver is fabricated in 28 nm CMOS and occupies 0.00547 mm^2 , as shown in [Fig. 37.9.7](#); synthesized digital logic is shared across all lanes. [Figure 37.9.5](#) shows the measured bathtub curve at 14 Gb/s, the recovered clock jitter histogram at 14 Gb/s, and jitter tolerance plot at 12 Gb/s and 14 Gb/s. The bathtub curve demonstrates that the proposed receiver achieves $<10^{-12}$ BER, with a timing margin of 0.295 UI. Moreover, with offset calibration, the TWPd locks at the optimal sampling point, yielding the largest margin. The measured jitter histogram exhibits 13.4ps peak-to-peak jitter and 1.982ps RMS jitter: jitter tolerance is measured for $<10^{-12}$ BER at 12 Gb/s for a comparison. High-frequency jitter tolerance (0.294 UI) is improved by 345% with background offset calibration, compared to no calibration (0.066 UI). At 14 Gb/s, the high-frequency jitter tolerance is measured to be 0.188 UI. Total power consumption is 2.28 mW at 14 Gb/s for an energy efficiency of 0.163 pJ/b. These results are achieved with compact per-lane hardware with a centralized controller to amortize its area overhead; the baud-rate phase tracking loop uses a low-resolution DCdL and simple decision logic, while the background offset-calibration loop adds only small capacitive trims and a few registers.

Figure 37.9.5

Measured recovered-data BER bathtub, recovered-clock jitter histogram at 14 Gb/s, and jitter tolerance plots.

[Figure 37.9.6](#) summarizes the results and compares the proposed DQ receiver with other memory interfaces: the proposed receiver achieves better energy efficiency and compact area due to its baud-rate phase-tracking loop and background offset calibration.

Figure 37.9.6
Performance summary and comparison to prior memory-interface receiver.



Figure 37.9.7
Chip micrograph with table summarizing area and power breakdown for each block.

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